THE CATEGORY OF I-ADICALLY COMPLETE MODULES OVER A NOETHERIAN RING HAS KERNELS AND COKERNELS BUT IS NOT ABELIAN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The closure argument uses the p-adic topology on the product; the delivered card L48284 (closed-is-complete) is the chapter's standing tool for closedness of kernels in complete modules.

Editorial note. The label lemma-complete-modules-not-abelian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 792-805 (section-non-abelian preamble, shared with delivered L48308): for I finitely generated the category A of I-adically complete R-modules has cokernel of phi: M -> N given by the I-adic completion of the usual cokernel (Algebra, hathat-finitely-generated) and kernel the usual kernel, which is closed and I-adically complete by Lemma closed-is-complete (delivered card L48284)
- Context span 841-859: with R = Z_p = lim Z/p^n Z and I = (p), the map diag(1, p, p^2, ...) from (direct sum of Z_p)^wedge to the product of Z_p sends (x_1, x_2, ...) with v_p(x_i) -> infinity to (x_1, p x_2, p^2 x_3, ...); the element (1, p, p^2, ...) lies in the closure of the image but not in the image; by the kernel/cokernel description this shows Im != Coim for this map

Source reference. Original definition 0317: sources/stacks-chow/tags.txt, source lines 3974–3974. Prerequisite pointer invoked at 875; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Prerequisite pointer invoked at 880; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0HB3: sources/stacks-examples/pinned-original-source.tex, source lines 808–808. Prerequisite pointer invoked at 882; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let R be a ring and let $I \subset R$ be a finitely generated ideal. Consider the category $\mathcal{A}$ of I-adically complete R-modules, see Algebra, Definition 0317. Let $\varphi : M \to N$ be a morphism of $\mathcal{A}$. The cokernel of φ in $\mathcal{A}$ is the I-adic completion $(\mathrm{Coker}(\varphi))^\wedge$ of the usual cokernel (as I is finitely generated this completion is complete, see Algebra, Lemma 05GG). Let $K = \mathrm{Ker}(\varphi)$; since φ is continuous this is a closed submodule of M. By Lemma 0HB3 below we see that K is I-adically complete and hence is the kernel of φ in $\mathcal{A}$.

We will give an example that shows that Im $\neq$ Coim in the category $\mathcal{A}$ in general. We take $R = \mathbf{Z}_p = \lim_n \mathbf{Z}/p^n\mathbf{Z}$ to be the ring of p-adic integers and we take $I = (p)$. Consider the map

$$\mathrm{diag}(1, p, p^2, \ldots) : \Big(\bigoplus\nolimits_{n\geq 1} \mathbf{Z}_p\Big)^\wedge \longrightarrow \prod\nolimits_{n\geq 1} \mathbf{Z}_p$$

where the left hand side is the p-adic completion of the direct sum. Hence an element of the left hand side is a vector $(x_1, x_2, x_3, \ldots)$ with $x_i \in \mathbf{Z}_p$ with p-adic valuation $v_p(x_i) \to \infty$ as $i \to \infty$. This maps to $(x_1, px_2, p^2x_3, \ldots)$. Hence we see that $(1, p, p^2, \ldots)$ is in the closure of the image but not in the image. By our description of kernels and cokernels above it is clear that Im $\neq$ Coim for this map.

Lemma 0.1. *Let R be a ring and let $I \subset R$ be a finitely generated ideal. The category of I-adically complete R-modules has kernels and cokernels but is not abelian in general even when R is Noetherian.*

Proof. See above. □